Deep Hedging under Black–Scholes and Heston Dynamics

by

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Contents

The hedge is learned through terminal portfolio error

$$\mathrm{HE} = \pi + \underbrace{\sum_{n=0}^{N-1} \delta_n \left(S_{t_{n+1}} - S_{t_n}\right)}_{\text{self-financing gains}} - (S_T - K)^+,$$

$$\min_{\pi,\theta} \mathbb{E}[\mathrm{HE}^2].$$

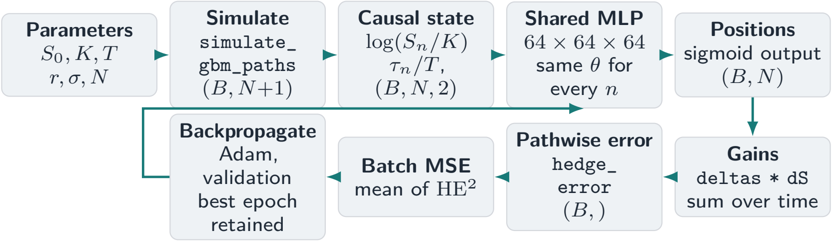
The analytic delta is never used as a training target.

- $N = 125$ discrete hedge dates in the baseline.
- $\delta_n = f_\theta(x_n)$ is adapted to information at t_n.
- Negative HE is a seller shortfall; reported seller loss is $L = -\mathrm{HE}$.
- With $r = 0$, the submitted experiments use undiscounted gains.

rebalance using current state

Source: Submitted report, Secs. 1.1, 2.2.2 and 3.1; Eqns. (1.1)–(1.2).

The implementation is differentiable end to end

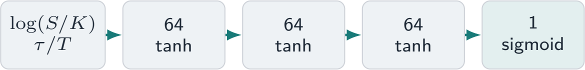


Supervision is financial: payoff minus self-financing wealth.

Boundary: Function names and path construction are verified from report Appendix C; the separate executable notebook bundle is absent from this repository.

Source: Submitted report, Algorithm A.6.1 and Listings C.1–C.3. Shape contract follows the displayed array construction and shared network input.

Call economics guide architecture selection



one shared map at all 125 dates

- **Log-moneyness** is scale-normalised and financially natural.
- **Weight sharing** exploits Markov and time-homogeneous structure.
- **Sigmoid** enforces the call-delta range $0 < \delta < 1$.
- A scalar premium is learned jointly with the hedge.

Source: Submitted report, Secs. 3.1.2–3.1.3; Tables 3.1 and A.3–A.6.

Candidate	Test RMSE	Time (s)
Shared norm. 64×3, tanh	**0.007891**	103.0
Shared norm. 64×2, ReLU	0.007912	93.2
Time-separated 16×1	0.008083	786.9

Selected on mean validation loss across three seeds; close alternatives remained.

Boundary: The model was selected, not uniquely dominant: the ReLU candidate won one seed.

The fair benchmark matches the grid and objective

No hedge
raw option risk

Black–Scholes delta
continuous-time rule sampled on the grid

Discrete-time MSE-optimal
conditional projection on the same grid

Degree-2 polynomial
deliberately low-capacity approximation

Neural hedge
terminal-MSE training without delta labels

same held-out paths

$$\delta_n^{\mathrm{DT}} = \frac{\mathrm{Cov}(g(S_T),\, \Delta S_n \mid S_{t_n})}{\mathrm{Var}(\Delta S_n \mid S_{t_n})}$$

The comparison asks whether the network recovers the finite-grid solution—not whether it beats correct Black–Scholes theory.

Source: Submitted report, Sec. 3.1.1 and Appendix A.4; Schweizer (1995, 2001).

The neural hedge recovers the finite-grid benchmark

Strategy	RMSE
No hedge	0.227 257
Black–Scholes delta	0.007 810
Discrete-time MSE-optimal	0.007 810
Degree-2 polynomial	0.022 183
Neural hedge	0.007 841

Boundary: These are final-benchmark values from an independent retraining, not architecture-search rows.

0.4%

neural RMSE above the discrete-time benchmark

0.163972

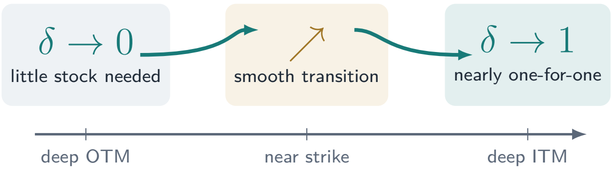
learned premium

vs. analytic 0.164111

Near equality is the validation result we wanted.

Source: Submitted report, Table 3.2 and Sec. 3.1.4.

The learned hedge has the right financial shape



- monotone in moneyness;
- close to both analytic curves;
- nearly indistinguishable on representative OTM, ATM and ITM paths.

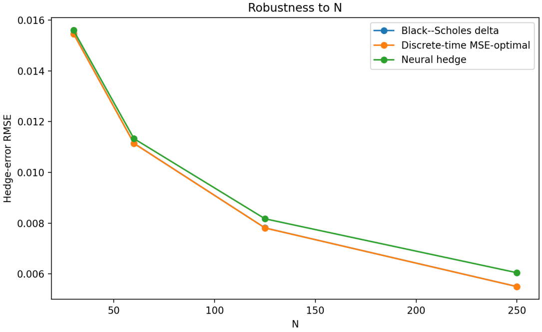
The network learned the expected hedge function, not only a low aggregate loss.

Source: Economic reading of submitted report Figures 3.3 and 3.5; schematic above is not a data redraw.

Parameter Robustness under Retraining

- The same architecture is retrained from scratch at each setting.
- Parameters are varied one at a time around the baseline $(K, \sigma, T, N) = (0.9,\ 0.4,\ 0.5,\ 125)$.

$K \in \{0.8, 0.9, 1.0, 1.1, 1.2\}$ $\sigma \in \{0.2, 0.4, 0.6\}$ $T \in \{0.25, 0.5, 1.0\}$ $N \in \{30, 60, 125, 250\}$



Mean neural-to-optimal RMSE ratio

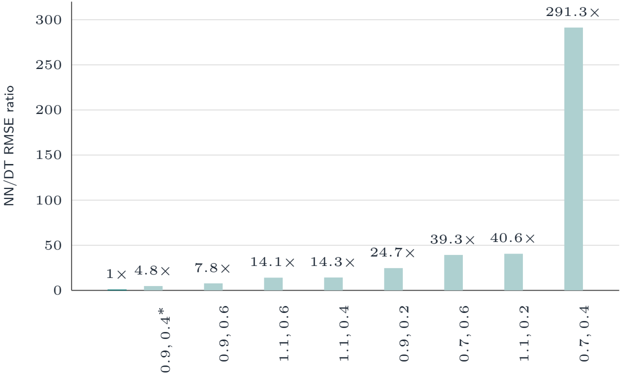
1.049

(Range 1.009–1.099)

- Neural hedge remains at most 10% above the discrete-time optimum
- Largest relative gap occurs at $N = 250$

Generalisation Failure of the Single-Scenario Hedger

- One model is trained at $(K^*, \sigma^*) = (0.9, 0.4)$ and evaluated elsewhere **without retraining**.



Horizontal axis: (K, σ) combinations.

Why does it fail?

1. **Single training scenario:** No variation from which to learn dependence on K or σ.
2. **Strike is hardcoded:** K^* is built into log-moneyness.
3. **Volatility is missing:** The network has no input through which it can respond to a change in σ.

The Parameter-Conditioned Hedger

	Original network	**Conditioned network**
Inputs	$\left(\log \frac{S}{K}, \frac{\tau}{T}\right)$	$\left(\log \frac{S}{K}, \frac{\tau}{T}, K, \sigma\right)$
Training	Fixed $K = 0.9,\ \sigma = 0.4$	$K \sim U(0.7, 1.1), \quad \sigma \sim U(0.2, 0.6)$
Paths	$100\,000$	$200\,000$
Premium	Learned jointly	Analytic Black–Scholes premium

Interpolation

Inside the training range

$$K \in [0.7, 1.1], \qquad \sigma \in [0.2, 0.6].$$

- Tested across nine parameter combinations.
- Maximum RMSE gap:$\leq$0.00018

Extrapolation

Outside the training range

$$K \in \{0.6, 1.3\}, \qquad \sigma \in \{0.1, 0.8\}.$$

- Strike extrapolation remains within 0.00004 of the optimum.
- Worst vol. gap at $\sigma = 0.1$: $\approx 3\times$ the optimum.

Submitted report, Sec. 4.1.3 and Table 4.3.

Heston adds a risk the stock cannot span

$$
\begin{aligned}
dS_t &= rS_t\,dt + \sqrt{v_t}S_t\,dW_t^S,\\
dv_t &= \kappa(\theta - v_t)\,dt + \xi\sqrt{v_t}\,dW_t^v,\\
dW_t^S\,dW_t^v &= \rho\,dt.
\end{aligned}
$$

$$dW_t^v = \rho\,dW_t^S + \sqrt{1-\rho^2}\,dW_t^\perp$$

- v_t is instantaneous variance; $\sqrt{v_t}$ is volatility.
- Since $|\rho| < 1$, an orthogonal variance shock remains.
- With $\rho = -0.70$, the stock cannot span the full variance shock.

Diffusion exposures

Instrument	dW^S	$dW^\perp$
stock	yes	no
liquid option	yes	yes

A liquid option with

$$\frac{\partial C^h}{\partial v} \neq 0$$

supplies a second locally independent exposure.

Scope: Local spanning is a continuous-time frictionless statement; discrete trading still leaves hedge error.

Source: Submitted report, Secs. 2.1.2 and 4.1.4; Heston (1993). Feller calculation in backup.

A liquid option supplies the missing variance exposure

▶ Neural hedge portfolio

Stock holding

δ_n: stock-shock exposure

$+$

Liquid-option holding

η_n: variance-sensitive exposure

Both positions are learned from terminal hedging error.

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10.

▶ Why the hedge option differs

	Target	Hedge
Strike	$K = 0.9$	$K_h = 1.0$
Maturity	$T = 0.5$	$T_h = 1.0$
Role	liability	traded hedge

- Initially at the money for useful variance sensitivity.
- Longer maturity so it remains alive throughout the target horizon.

Each training sample is a full Heston trajectory

independent shocks ► **correlate** ► **update state** ► **price hedge option** ► **form state** ► **choose positions**

► **Simulation**

Log-Euler stock and truncation-based variance update.

► **Grid**

125

rebalancing intervals over six months

► **Data split**

100k training

25k validation

100k fresh test trajectories

Boundary: Full-information experiment: current simulated variance enters the state.

Source: Submitted report, Listing C.4 and Algorithms A.3.3, A.6.8, A.6.10.

A tempting shortcut: freeze the current volatility

$$C_t^h \approx C_{\mathrm{BS}}(S_t, K_h, T_h - t, \sqrt{v_t})\,.$$

at t_n: insert $\sigma = \sqrt{v_n}$ → price as if σ stays fixed to T_h → at t_{n+1}: repeat with new variance

A plausible price level is not enough: the whole price path must be Heston-consistent.

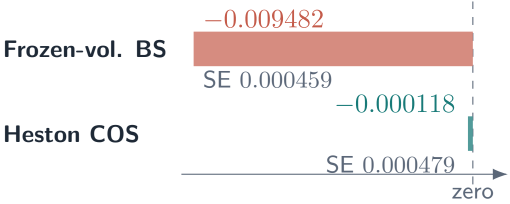
Source: Submitted report, Sec. 4.1.4 (pp. 24–25) and Algorithm A.6.5.

- **Why it seemed reasonable**
  - cheap to evaluate;
  - responsive to current variance;
  - creates a variance-sensitive second instrument.

Question Does repeatedly recomputing the Black–Scholes level produce a risk-neutral martingale under the simulated Heston dynamics?

The martingale test catches the inconsistency

Mean total liquid-option movement over $[0, T]$; the martingale reference is zero.



▶ **Proxy signal**

-20.6

standard errors below zero

COS movement is not statistically distinguishable from zero at ordinary Monte Carlo precision.

The proxy is not just imprecise in level; its path dynamics are inconsistent with the simulated Heston market.

Source: Submitted report, Table A.9 and Sec. 4.1.4. Diagnostic evidence, not a standalone arbitrage proof.

Why the frozen-volatility proxy drifts

Frozen-volatility liquid hedge-option proxy: $g(t, S, v) = C_{\mathrm{BS}}\big(S, K_h, T_h - t, \sqrt{v}\big)$

Apply Heston dynamics to the Black–Scholes proxy using Itô's lemma.

- **Cancelled by the frozen-v Black–Scholes PDE**

$$g_t + rSg_S + \frac{1}{2}vS^2 g_{SS} - rg = 0$$

- **Residual drift under Heston**

$$R(t, S, v) = \kappa(\theta - v)g_v + \frac{1}{2}\xi^2 v g_{vv} + \rho\xi v S g_{Sv}$$

- **1** variance mean reversion
- **2** variance curvature
- **3** spot–variance interaction

Repricing the Black–Scholes level at every date does not make the resulting option-price process satisfy the Heston pricing equation.

Scope: The residual is state-dependent; under the submitted simulation its average contribution was negative.

Source: Submitted report, Sec. 4.1.4 and App. A.6.3; Black–Scholes (1973); Heston (1993).

COS evaluates a Heston-consistent option price

Pricing and marking

Monte Carlo Heston states → **COS option prices**

COS uses the Heston characteristic function to mark the liquid option consistently with each simulated state.

Position selection

Analytic local COS benchmark

Option matches variance exposure; stock removes the remaining delta exposure.

$$\eta_n = \frac{V_n^T}{V_n^h}, \qquad \delta_n = \Delta_n^T - \eta_n \Delta_n^h$$

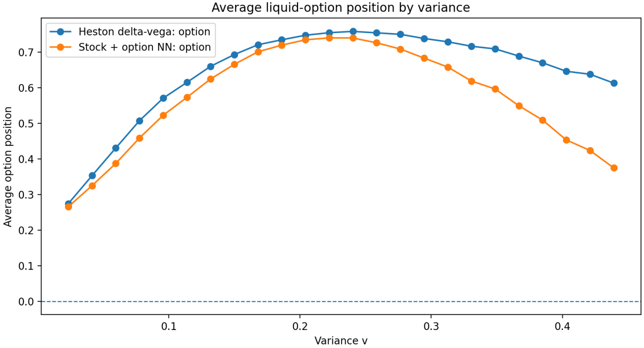
Neural policy

Stock and option positions are learned directly from terminal hedging error; **COS Greeks are not training labels**.

Same traded COS option path; different position rule.

Source: Submitted report, Sec. 4.1.4 and Apps. A.6.1–A.6.2; Heston (1993); Fang & Oosterlee (2008).

The learned option position adapts to the variance state



The stock-plus-option network broadly follows the local Heston COS delta–vega option position, while using less aggressive positions at higher variance.

Source: Submitted report, Figure 4.3 and Sec. 4.1.4. Vector crop from the final report; original export was not present in the submitted presentation repository.

The stock-plus-option NN lowers hedge error in every submitted seed

Representative full-information run

Strategy	RMSE	Loss CVaR95
No hedge	0.190 061	0.492 959
BS-proxy stock delta	0.019 997	0.046 360
Heston COS delta–vega	0.014 782	0.033 251
Stock-only NN	0.018 166	0.041 379
BS-proxy Greek heuristic on the COS option path	0.009 874	0.023 130
Stock-plus-option NN	0.007 426	0.016 624

Lower RMSE in all 3 submitted seeds

22.7%
mean RMSE improvement

27.1%
mean Loss CVaR95 improvement

Heston COS delta–vega is the model-consistent local benchmark; the BS-proxy Greek rule on the COS path is the strongest tested analytic heuristic.

Fair-centred separately: compares residual hedge dispersion and tail risk.

Different objectives: local Greek matching versus finite-grid terminal-error training.

Source: Submitted report, Tables 4.5–4.6 and A.11. Loss CVaR95 is mean seller loss in the worst 5% of test outcomes.

Three conclusions survive the diagnostics

▶ **What the experiments establish**

1. **Black–Scholes:** the neural strategy recovered the known finite-grid hedge.
2. **Conditioning:** adding the missing contract information repaired the identified generalisation failure.
3. **Heston:** with a consistently priced liquid option, the stock-plus-option NN achieved lower terminal error than the strongest tested analytic Greek heuristic in every submitted seed.

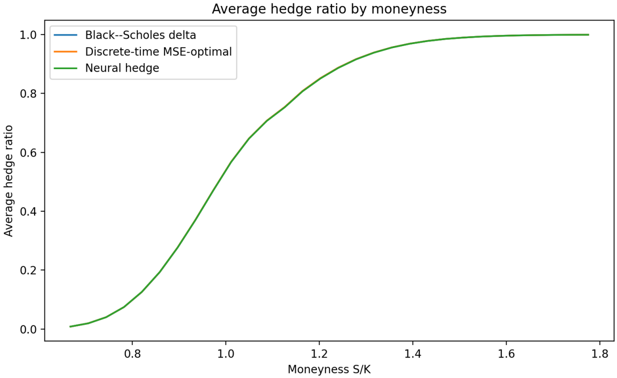
▶ **What they do not establish**

- high-powered statistical evidence beyond the three submitted seeds;
- a derived finite-grid optimal stock-and-option hedge;
- performance beyond the frictionless simulated Heston setting and submitted information assumptions.

A constrained neural policy can recover a known finite-grid solution and then be trained directly against the finite-grid terminal objective as the dynamics and traded instrument set become more complex.

Source: Submitted report, Secs. 4.1.4 and 5; stated limitations.

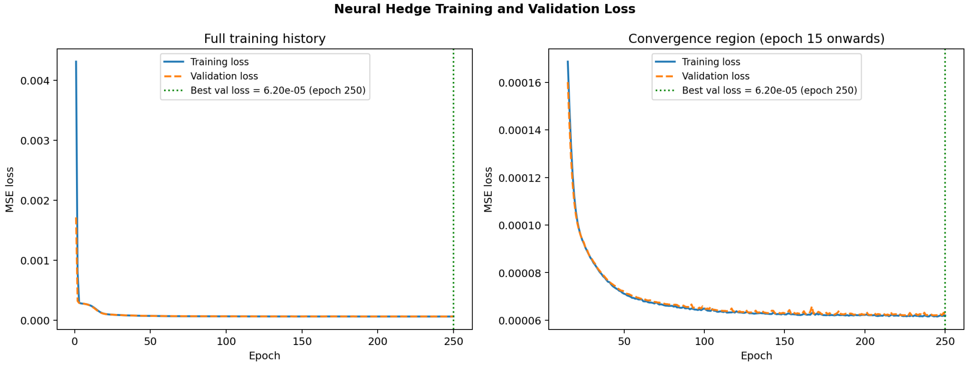
Backup: the learned Black–Scholes hedge matches the analytic curves



The learned hedge ratio closely follows the Black–Scholes delta and discrete-time MSE-optimal curves across moneyness.

Source: Submitted report, Figure 3.3 and Sec. 3.1.4. Vector crop from the final report.

Backup: training and validation losses converge without visible instability



Training and validation losses track closely and flatten late in training; the best validation loss occurs at the submitted epoch cap of 250.

Source: Submitted report, Figure 3.6 and Sec. 3.1.4. Optimisation and validation evidence, not proof of external generalisation.

State and tensor shapes

BS paths $(B, N+1)$ $\to$ state stack $(B, N, 2)$ $\to$ shared calls $(B \cdot N, 2)$ $\to$ deltas (B, N) $\to$ terminal error $(B,)$

Heston S, v, C^h each $(B, N+1)$ $\to$ state stack $(B, N, 6)$ $\to$ shared calls $(B \cdot N, 6)$ $\to$ (δ, η) $(B, N, 2)$ $\to$ gain / error $(B,)$

The same shared network is applied to every path-date state. Each complete path produces one terminal gain and one terminal hedge error.

Source: Submitted report, Listings C.1–C.4 and Algorithm A.6.10.

How gradients accumulate through the shared hedge

$$(\delta_{i,n}, \eta_{i,n}) = f_w(x_{i,n}), \qquad L = B^{-1} \textstyle\sum_{i=1}^{B} HE_i^2.$$

$$HE_i = \pi_w + \sum_{n=0}^{N-1} [\delta_{i,n} \Delta S_{i,n} + \eta_{i,n} \Delta C^h_{i,n}] - H_i.$$

$$\boxed{\nabla_w L = \frac{2}{B} \sum_{i=1}^{B} HE_i \sum_{n=0}^{N-1} \left[\Delta S_{i,n} \nabla_w \delta_{i,n} + \Delta C^h_{i,n} \nabla_w \eta_{i,n}\right]}$$

$$L \leftarrow HE \leftarrow G_T \leftarrow (\delta, \eta) \leftarrow f_w \leftarrow w$$

- one weight vector w is reused at every date;
- gradients accumulate across all paths and dates;
- no analytic delta or vega labels are supplied;
- COS prices are precomputed marks, outside the gradient path.

Source: Submitted report, Sec. 2.2.2 and Algorithm A.6.10.

Heston simulation

- **Correlated shocks**

$$Z_n^S, Z_n^{\perp} \sim N(0,1), \qquad Z_n^v = \rho Z_n^S + \sqrt{1-\rho^2}\, Z_n^{\perp}.$$

- **Log-Euler stock update**

$$S_{n+1} = S_n \exp\left[\left(r - \tfrac{1}{2}v_n^+\right)\Delta t + \sqrt{v_n^+ \Delta t}\, Z_n^S\right].$$

Preserves $S_{n+1} > 0$ and respects multiplicative stock dynamics.

- **Truncation-based variance update**

$$\widetilde{v}_{n+1} = v_n + \kappa(\theta - v_n^+)\Delta t + \xi\sqrt{v_n^+ \Delta t}\, Z_n^v,$$
$$v_{n+1} = \max(\widetilde{v}_{n+1}, 0), \qquad v_n^+ = \max(v_n, 0).$$

Ordinary Euler–Maruyama can propose negative variance, making the next square root invalid.

Boundary: Standard Monte Carlo generates many paths; log-Euler and full truncation specify each step. This is not an exact Heston transition. Feller positivity does not prevent negative Euler proposals. Discrete paths versus continuous-time COS marks contribute to the small residual COS drift.

Source: Heston (1993); Lord, Koekkoek & van Dijk (2010); submitted report, Alg. A.3.3 and Listing C.4.

Two-instrument gain construction

▶ **Causal state and shared policy**

$$x_n = \left(\log \frac{S_n}{K}, \frac{T - t_n}{T}, \frac{v_n}{\theta}, \log \frac{S_n}{K_h}, \frac{T_h - t_n}{T_h}, \frac{C_n^h}{S_n}\right)$$

$$(\delta_n, \eta_n) = 5 \tanh\big(f_w(x_n)\big).$$

▶ **Terminal gain and training loss**

$$G_T = \sum_{n=0}^{N-1} \left(\delta_n \Delta S_n + \eta_n \Delta C_n^h\right),$$

$$HE = \pi_w + G_T - (S_T - K)^+,$$

$$L = \mathbb{E}[HE^2].$$

Boundary: Executable notebooks and validation outputs were submitted separately; this slide maps the presentation equations to the submitted report algorithms and listings.

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10.

Holdings, trades, and gains

- **Network action: total holdings**

$$(\delta_n, \eta_n) \quad \text{over } [t_n, t_{n+1}].$$

- **Trade increments**

$$\Delta\delta_n = \delta_n - \delta_{n-1}, \qquad \Delta\eta_n = \eta_n - \eta_{n-1}.$$

Source: Submitted report, Sec. 4.1.4 and Algorithm A.6.10.

- **Frictionless one-step gain**

$$\delta_n \Delta S_n + \eta_n \Delta C_n^h.$$

- the submitted gain equation uses holdings;
- trade increments determine transaction costs and turnover;
- calling δ_n or η_n a trade changes the implementation's meaning.

Why the Heston outputs use scaled tanh

- **Two signed outputs**

$$\delta_n = 5\tanh(z_{1,n}), \qquad \eta_n = 5\tanh(z_{2,n}).$$

$$[-5, 5]$$

numerical output range

- **Why this map is used**
  - stock and option holdings can be long or short;
  - sigmoid's $(0, 1)$ range is unsuitable;
  - scaled tanh permits signed, bounded positions;
  - the bound improves numerical stability, but is not an optimality theorem.

Submitted diagnostic: $\max|\eta^{NN}| \approx 1.33$, comfortably below five.

Source: Submitted report, Sec. 4.1.4, Algorithm A.6.10 and Table A.12.

Frozen-volatility drift derivation

For the liquid hedge-option proxy $g(t, S, v) = C_{\mathrm{BS}}(S, K_h, T_h - t, \sqrt{v})$: $(dS)^2 = vS^2dt, \quad (dv)^2 = \xi^2 v dt, \quad dS\,dv = \rho\xi vSdt.$

$$dg = \left[g_t + rSg_S + \kappa(\theta - v)g_v + \tfrac{1}{2}vS^2 g_{SS} + \tfrac{1}{2}\xi^2 v g_{vv} + \rho\xi vSg_{Sv}\right]dt$$
$$+ g_S S\sqrt{v}\,dW^S + g_v \xi\sqrt{v}\,dW^v.$$

▶ **Frozen-v PDE cancellation**

$$g_t + rSg_S + \tfrac{1}{2}vS^2 g_{SS} - rg = 0$$

▶ **Residual discounted drift**

$$\kappa(\theta - v)g_v + \tfrac{1}{2}\xi^2 v g_{vv} + \rho\xi vSg_{Sv}$$

Boundary: At submitted $r = 0$, discounted and undiscounted residuals coincide. The sign is state-dependent; this diagnostic is not a standalone arbitrage proof.

Source: Submitted report, Sec. 4.1.4 and App. A.6.3; Black–Scholes (1973); Heston (1993).

Heston characteristic function used by COS

- **Conditional transform of log price**

$$\phi(u;\tau,x,v) = \mathbb{E}^{\mathbb{Q}}\left[e^{iuX_T} \mid X_t = x, v_t = v\right] = \exp(C(u,\tau) + D(u,\tau)v + iux)\,, \qquad X_T = \log S_T.$$

$$u_k = \frac{k\pi}{b-a}, \qquad k = 0,\ldots,N_{\mathrm{COS}} - 1.$$

- **1** ϕ encodes the conditional distribution in transform space.
- **2** COS samples ϕ on a deterministic frequency grid.
- **3** The finite sum gives price, delta and $\partial C/\partial v$.

Source: Heston (1993); Fang & Oosterlee (2008); submitted report, Algorithm A.6.6.

COS pricing

Deterministic semi-analytical pricing: no nested Monte Carlo inside COS.

$$C_{\mathrm{COS}} \approx e^{-r\tau} \sum_{k=0}^{N_{\mathrm{COS}}-1}{}' A_k V_k,$$

$$A_k = \frac{2}{b-a} \Re\left[\phi(u_k) e^{-iu_k a}\right], \qquad u_k = \frac{k\pi}{b-a}.$$

The prime halves $k = 0$. A_k is a density coefficient; V_k is the analytic payoff coefficient. u_k is the k-th cosine frequency on $[a, b]$.

Heston supplies $\phi(u)$; COS evaluates the conditional expectation from $\phi(u_k)$.

- $N_{\mathrm{COS}} = 256$, $[a, b] = [-7, 7]$;
- traded range $\tau_h \in [0.5, 1]$;
- Carr–Madan: $\alpha = 1.5$, $u \in [0, 200]$, 20,000 points;
- price discrepancy 5.1×10^{-8};
- finite-difference errors: delta 9.8×10^{-5}, variance sensitivity 4.7×10^{-4}.

Boundary: Validation is local to the submitted traded range.

Source: Heston (1993); Fang & Oosterlee (2008); Carr & Madan (1999); submitted report, Apps. A.6.1–A.6.2.

Analytic Heston stock-and-option hedge

$$dC^T = \cdots + \Delta^T S\sqrt{v}\,dW^S + V^T \xi\sqrt{v}\,dW^v,$$
$$d(\delta S + \eta C^h) = \cdots + (\delta + \eta\Delta^h)S\sqrt{v}\,dW^S + \eta V^h \xi\sqrt{v}\,dW^v.$$

Match $\eta V^h = V^T$ and $\delta + \eta\Delta^h = \Delta^T$: $\boxed{\eta = \frac{V^T}{V^h}, \qquad \delta = \Delta^T - \eta\Delta^h}$

- **Heston COS delta–variance benchmark** Heston COS supplies target and liquid hedge-option sensitivities.
- **BS-proxy Greek heuristic on the COS option path** Frozen-v BS sensitivities set positions; gains use the same COS marks.

Boundary: Local continuous-time first-order diffusion matching; not the finite-grid terminal-MSE optimum. The report table retains "delta–vega", although $V = \partial C/\partial v$.

Source: Submitted report, Sec. 4.1.4 and Algorithms A.6.8–A.6.9.

Pricing accuracy and finite-grid hedge optimality are different questions

- **What COS establishes**
  - model-consistent Heston option prices;
  - independently validated prices and sensitivities;
  - local first-order exposures for the COS Greek rule.
- **What it does not optimise**
  - finite-grid terminal MSE;
  - between-date gamma and variance curvature;
  - spot–variance cross effects and changing Greeks.

Model misspecification may accidentally compensate for finite-grid effects in this tested parameter regime. That ordering does **not** imply that Black–Scholes is a better Heston pricer.

The neural policy trains directly on terminal MSE. Turnover is supporting diagnostic evidence, not proof of the ordering.

Source: Submitted report, Sec. 4.1.4 and Tables 4.5–4.6.

Fair-premium centring

$$\boxed{\pi_j^* = \arg\min_{\pi} \mathbb{E}_{\text{test}}\left[(\pi + G_T^j - H)^2\right] = \mathbb{E}_{\text{test}}[H - G_T^j]}$$

$$HE_j = \pi_j^* + G_T^j - H, \qquad \mathbb{E}_{\text{test}}[HE_j] = 0.$$

- applied separately to every headline strategy;
- removes test-sample mean error;
- RMSE compares centred dispersion; Loss CVaR95 compares centred downside tail risk;
- not used during training: the NN jointly learns a raw premium and hedge weights.

Source: Submitted report, Sec. 4.1.4, Tables 4.5–4.6 and A.11, and Algorithm A.6.10.

Data roles remain separate

Training	Fit weights and learned premium
Validation	Select architecture / best epoch
Test	Final fair-centred comparison

Boundary: Because π_j^* is estimated on the same test paths as the metrics, this is not fully out-of-sample joint pricing and hedging.

Three-seed Heston comparison

- **Comparator: BS-proxy Greek heuristic on the COS option path**

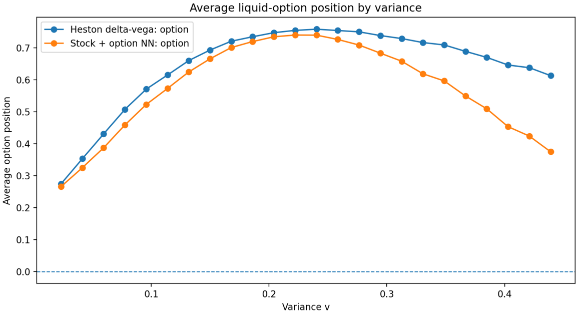
Seed	NN RMSE	Heuristic RMSE	RMSE imp. (%)	CVaR95 imp. (%)
111	0.007 806	0.009 838	20.6	26.3
222	0.007 715	0.009 871	21.8	27.0
333	0.007 346	0.009 864	25.5	28.0
	Mean pairwise improvement		22.7	27.1

- The three-seed cell retrained the stock-only tanh and stock-plus-option NNs and reevaluated the COS delta–variance and BS-proxy Greek analytic rules on each fresh test set.
- No hedge, BS-proxy stock delta and representative-only rows in the main table were not part of this four-strategy rerun.

Boundary: Three seeds demonstrate consistency across submitted runs; they do not support high-powered confidence intervals.

Source: Submitted report, Table A.11.

Backup: learned option position in the representative Heston run



▶ **Exact setting** Representative full-information COS run; horizontal axis is the Heston variance state (v), and the vertical axis is average liquid-option position.

▶ **Defensive interpretation** The NN broadly follows the local Greek hedge but softens at high variance, consistent with optimising terminal MSE rather than copying the analytic rule.

Source: Submitted report, Figure 4.3 and Sec. 4.1.4. In this run, $\max|\eta^{\mathrm{NN}}| \approx 1.33$, inside the imposed $[-5, 5]$ range.

Observable-information robustness

- **Causal variance proxy**
  - EWMA decay $\lambda = 0.94$;
  - $\mathrm{corr}(\hat{v}_t, v_t) = 0.777613$;
  - relative variance-proxy RMSE $= 0.369375$.

- **Submitted mean RMSE**

Strategy	RMSE
Stock-plus-option NN	$0.007\,877$
BS-proxy Greek heuristic on the COS option path	$0.010\,734$

The EWMA state is a causal compressed history of past squared returns; the MLP itself remains non-recurrent.

Boundary: C_t^h/S_t remains contemporaneously observable and variance-informative, so this is not a stock-return-only filtering experiment.

Source: Submitted report, Sec. 4.1.4 and Tables A.10, A.12, A.15; J.P. Morgan/Reuters (1996).

Why the hedge option differs from the target

- **Target liability**

$$K = 0.9, \quad T = 0.5$$

- **Liquid hedging option**

$$K_h = 1, \quad T_h = 1$$

- near-ATM options generally have useful variance sensitivity;
- $T_h > T$ keeps the liquid hedging option alive throughout the target horizon;
- trading an identical target option would largely trivialise the problem;
- the target liability need not itself be a liquid traded contract.

Source: Submitted report, Sec. 4.1.4 and the target/hedge-option configuration.

What the martingale diagnostic measures

- **Sample mean over $[0, T]$**

$$\widehat{m} = \frac{1}{M}\sum_{i=1}^{M}\left(C_T^{h,(i)} - C_0^h\right).$$

- **Monte Carlo standard error**

$$\widehat{\mathrm{SE}}(\widehat{m}) = \frac{s(C_T^h - C_0^h)}{\sqrt{M}}.$$

- individual paths fluctuate; the statistic concerns average movement;
- under $r = 0$, the reference is zero;
- the target horizon ends at $T = 0.5 < T_h = 1$;
- this diagnoses dynamic inconsistency, not standalone executable arbitrage.

Boundary: Proxy: -0.009482 (SE 0.000459); COS: -0.000118 (SE 0.000479). The COS residual is negligible, not exactly zero.

Source: Submitted report, Table A.9 and Algorithm A.6.8.

Backup: the discrete-time conditional projection

$$\min_{\pi,\delta} \quad \mathbb{E}\left[\left(\pi + \sum_{j=0}^{N-1} \delta_j \Delta S_j - H\right)^2\right],$$

$$0 = \mathbb{E}\left[\left(H - \pi - \sum_{j=0}^{N-1} \delta_j \Delta S_j\right) \Delta S_n \mid \mathcal{F}_{t_n}\right].$$

Under submitted $r = 0$, S is a martingale, so

$$\delta_n^{\mathrm{DT}} = \frac{\mathbb{E}[H \Delta S_n \mid \mathcal{F}_{t_n}]}{\mathbb{E}[(\Delta S_n)^2 \mid \mathcal{F}_{t_n}]} = \frac{\mathrm{Cov}(H, \Delta S_n \mid S_{t_n})}{\mathrm{Var}(\Delta S_n \mid S_{t_n})}.$$

Boundary: This simplification is tied to the submitted $r = 0$ martingale setting; local and global quadratic hedges need not coincide more generally.

Source: Submitted report, Appendix A.4; Schweizer (1995, 2001).

Backup: method-to-code map

Mathematical concept	Displayed implementation	Report evidence	Primary expert
Exact BS path simulation	`simulate_gbm_paths`	Listing C.1; Alg. A.3.1	Micaela / Glasson
Self-financing terminal error	`hedge_error`	Listing C.2; Alg. A.6.1	Micaela / Glasson
Shared Markov hedge	`SharedMarkovHedge`	Listing C.3; Sec. 2.2.2	Kaiden / Micaela
Heston full-truncation simulation	PyTorch loop in Listing C.4	Listing C.4; Alg. A.3.3	Glasson / Siphelele
COS liquid-option pricing	Algorithms A.6.6–A.6.8	App. A.6.1–A.6.2	Glasson
Two-instrument neural gain	Algorithm A.6.10	Sec. 4.1.4	Siphelele / Glasson

Boundary: Executable notebooks and validation outputs were submitted separately; this slide maps the presentation equations to the submitted report algorithms and listings.

Backup: transaction costs change the learned policy

$$V_T^{\lambda} = \pi + \sum_{n=0}^{N-1} \delta_n \Delta S_n - \lambda \sum_{n=0}^{N-1} S_{t_n} |\delta_n - \delta_{n-1}|, \qquad \delta_{-1} = 0.$$

λ	Strategy	RMSE	Loss CVaR95	Turnover
0.0025	BS cost-adjusted	0.008 492	0.021 464	4.497 364
0.0025	Cost-aware NN	0.008 077	0.018 694	4.386 880
0.0050	BS cost-adjusted	0.010 013	0.025 565	4.497 364
0.0050	Cost-aware NN	0.008 331	0.019 269	4.284 602

Boundary: Comparator is a cost-adjusted frictionless delta, not a full optimal-cost benchmark such as Leland or Whalley–Wilmott.

Source: Submitted report, Appendix A.7 and Table A.16.

Backup: data-generation comparison

Paths	Method	RMSE	Loss CVaR95
10,000	Crude MC	0.009 430	0.021 266
10,000	LHS	0.009 201	0.020 845
10,000	Sobol + Brownian bridge	0.009 174	0.020 865
100,000	Crude MC	0.007 865	0.018 218
100,000	LHS	0.007 846	0.018 331
100,000	Sobol + Brownian bridge	0.007 840	0.018 125

At 10,000 paths, structured methods reduce RMSE by roughly 2.4–2.7%; by 100,000 paths the difference is negligible.

Source: Submitted report, Sec. 3.1.5 and Tables A.7–A.8.

Backup: experimental protocol

Experiment	Train	Val	Test	Batch	Training / seeds
Architecture selection	100k	20k	50k	256	60 epochs; 2026–2028
Final BS benchmark	100k	30k	100k	4096	250 epochs; base 123
Parameter robustness	50k	15k	50k	4096	150 epochs
Conditioned hedger	200k	30k	50k / case	4096	250 epochs; 777
Heston stock + option	100k	25k	100k	4096	150 epochs; base 123

- Adam learning rate 10^{-3} unless otherwise stated.
- Baseline: $S_0 = 1, K = 0.9, T = 0.5, r = 0, \sigma = 0.4, N = 125.$
- Architecture search uses more updates per epoch via batch size 256; final experiments use 4096.

Source: Submitted report, Appendix A.1, Table A.1.

Backup: AI assistance and verification

Assistance area disclosed	Human verification reported
Code drafting and debugging	Executed notebooks, code inspection and exported results
Mathematical explanations and LaTeX	Checked against theory and cited sources
Report organisation and consistency	Tables checked against CSV/TeX outputs and figures
COS implementation support	Independent Carr–Madan Fourier price check
Neural experiment support	BS analytic and finite-grid benchmarks; archived outputs
Documentation and reproducibility	Checklists, packaging review and author decisions

Boundary: The report names ChatGPT and Claude. It states that AI suggestions were not authoritative and that all final methodological and interpretive decisions remained with the authors.

Source: Submitted report, Appendix B.

Backup: architecture selection by seed

Seed	Validation winner	Val. loss	Test RMSE
2026	Shared norm. 64×3, tanh	6.025×10^{-5}	$0.007\,883$
2027	Shared norm. 64×3, tanh	6.164×10^{-5}	$0.007\,916$
2028	Shared norm. 64×2, ReLU	6.014×10^{-5}	$0.007\,855$

The 64×3 tanh model won two seeds and had the best mean validation loss and test RMSE; the ReLU alternative was close.

Source: Submitted report, Tables A.3–A.6.

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